

Deep Learning

Ex 2 Theoretical

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1. Q1

(1 Mark)

Show that a convolution with respect to any filter h is time/space invariant.

Ans:

Convolving with filter h is defined as:

$$y(t) = (x * h)(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau$$

shifting the input will yield $x'(t) = x(t - t_0)$, hence the output is:

$$y'(t) = (x' * h)(t) = \int_{-\infty}^{\infty} x'(\tau)h(t - \tau) d\tau = \int_{-\infty}^{\infty} x(\tau - t_0)h(t - \tau) d\tau$$

Now, substitute $\tau' = \tau - t_0$, gives:

$$y'(t) = \int_{-\infty}^{\infty} x(\tau')h((t - t_0) - \tau') d\tau' = (x * h)(t - t_0) = y(t - t_0)$$

2. Q2

(4 Marks)

Explain whether each of the following layers are time/space invariant or not:

- (a) Additive constant
- (b) Pointwise nonlinearity (such as ReLU)
- (c) Strided pooling by a factor
- (d) As a result, is a CNN composed of all these operators (+convolution) time invariant?

(4 Marks)

Ans:

- (a) Adding a constant c to the input is time/space invariant. If the input is shifted, the same constant is still added, resulting in the same shift in the output. i.e., for a system defined as:

$$y(t) = x(t) + c$$

where c is a constant, if we delay the input by d to get $x(t - d)$, the output becomes:

$$y_d(t) = x(t - d) + c = y(t - d)$$

- (b) Pointwise nonlinearity is time/space invariant. If the input is shifted, the function works on the same input, but shifted, making it the image of the function on the shifted input. i.e., For a pointwise nonlinearity f , where:

$$y(t) = f(x(t))$$

When the input is delayed by d , we get:

$$y_d(t) = f(x(t - d)) = y(t - d)$$

- (c) Strided pooling however, is not time/space invariant. suppose we take a maxpool in a window of 2 by 2, on the next image $[0, 0, 0, 0][0, 255, 255, 0][0, 255, 255, 0][0, 0, 0, 0]$. (if out of range, pad with zeros) then starting from index 0, will yield $[255, 255][255, 255]$, and starting from index 1 will yield, $[255, 0][255, 0]$, making the output dependent on time and space.
- (d) Taking into account what we have proven in the last parts, a CNN composed of convolutions, additive constants, pointwise nonlinearity, and strided pooling by a factor is not time/space invariant. removing the pooling layer however, will make it time/space invariant as composition of time/space invariant functions.

3. Q3

(3 marks)

Calculate the Jacobian matrix of the following layers:

- (a) Additive bias vector
- (b) General matrix multiplication
- (c) Convolution layer

(3 marks)

Ans:

(a) For a layer defined as:

$$y = x + b$$

where $x \in \mathbb{R}^n$ and $b \in \mathbb{R}^n$ are vectors, the output is given by:

$$y_i = x_i + b_i$$

Calculating the Jacobian:

$$J_{ij} = \frac{\partial y_i}{\partial x_j} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

which means

$$J = I_n$$

(b) For a fully connected layer:

$$y = Wx$$

where $W \in \mathbb{R}^{m \times n}$, the output is given by:

$$y_i = \sum_{j=1}^n W_{ij} x_j$$

Calculating the Jacobian:

$$J_{ij} = \frac{\partial y_i}{\partial x_j} = W_{ij}$$

which means

$$J = W$$

(c) Suppose the convolution is 1D with an input x of length n and a filter h of length k , The output is:

$$y_i = \sum_{j=0}^{k-1} h_j \cdot x_{i+j}$$

Calculating the Jacobian:

$$\frac{\partial y_i}{\partial x_j} = \begin{cases} h_{j-i} & \text{if } 0 \leq j - i < k \\ 0 & \text{otherwise} \end{cases}$$

which means

$$J = \begin{pmatrix} h_0 & h_1 & \cdots & h_{k-1} & 0 & \cdots & 0 \\ 0 & h_0 & h_1 & \cdots & h_{k-1} & 0 & \cdots \\ \vdots & & \ddots & & & \ddots & \\ 0 & \cdots & 0 & h_0 & h_1 & \cdots & h_{k-1} \end{pmatrix}$$